淘宝用户行为数据分析报告

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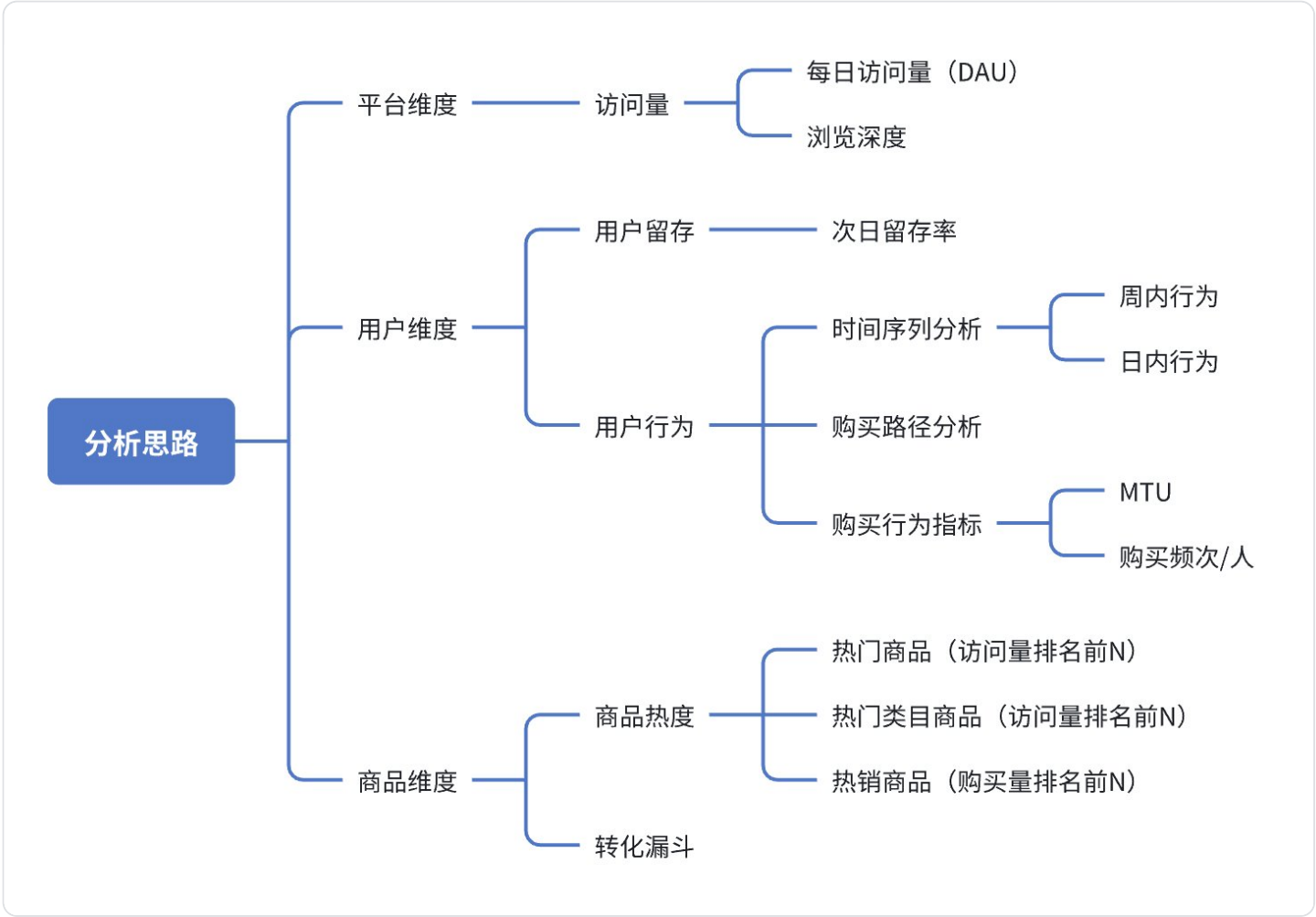
前言

数据集介绍

- **数据来源**：源自阿里云天池的淘宝平台用户行为日志数据，经筛选处理后约999k条（见Appendix）
- **数据内容**：
  - `user_id`（用户 ID），共计527929个用户；
  - `item_id`（商品 ID），共计479307件商品；
  - `category_id`（品类 ID），共计6407种品类；
  - `behavior_type`（行为类型），包含 `pv` - 浏览、`fav` - 收藏、`cart` - 加购、`buy` - 购买；
  - `timestamps`（时间戳），经处理后拆为具体时间、小时和日期。
- **处理后的数据示例**：

id	user_id	item_id	category_id	behavior_type	timestamps	datetimes	dates	times	hours
24	1	4092065	2355072	pv	1511910242	2017-11-29 07:04:02	2017-11-29	07:04:02	07
51	1	4954999	411153	pv	1512264993	2017-12-03 09:36:33	2017-12-03	09:36:33	09
223	1000	1585986	1299190	pv	1512230625	2017-12-03 00:03:45	2017-12-03	00:03:45	00
244	1000001	2999487	1973012	pv	1511924532	2017-11-29 11:02:12	2017-11-29	11:02:12	11
296	1000004	40945	4756105	pv	1511703467	2017-11-26 21:37:47	2017-11-26	21:37:47	21
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

分析思路

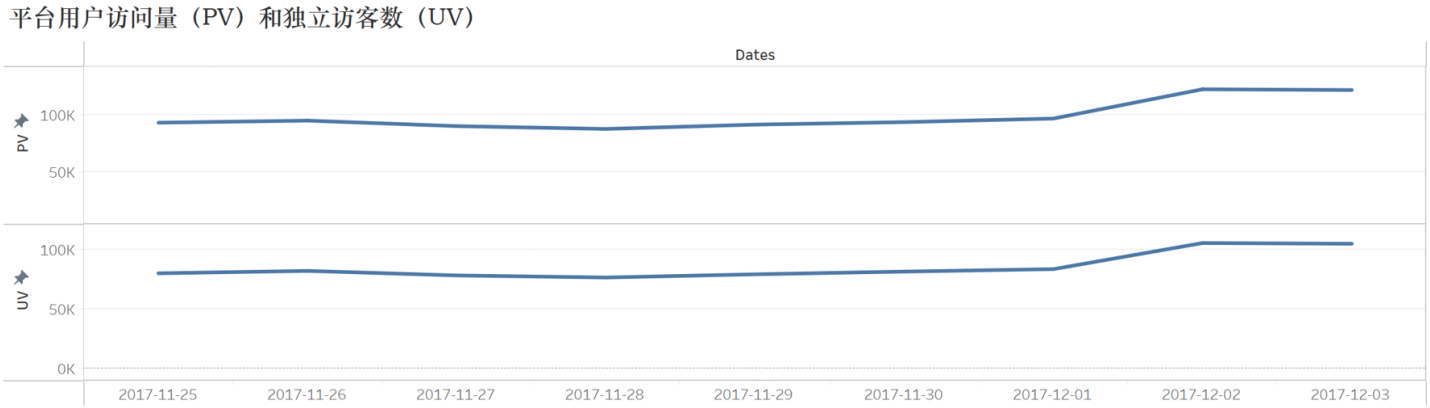


报告正文

一、平台流量监测体系（平台维度）

1.1 基础流量指标

- **核心指标**：
  - `PV`：总访问量，衡量每日平台流量规模
  - `UV`：独立访客数（可理解为DAU），衡量每日平台活跃用户数
  - `PV/UV`：浏览深度，衡量用户参与度/活跃度
- **数据结果：**
  - **每日访问量先降后升：**每日访问量总体呈现先下降后上升的趋势，2017-12-01后明显增长，可能受双十二活动驱动影响。
  - **平台的用户参与度有待提高：**数据统计期间（2017-11-25至2017-12-03）总独立访客数为493k，而每日平均独立访客数在100k左右，每日独立访客渗透率不到1/4，且每日访问深度（pv/uv）稳定在1.2次左右，因此平台的用户参与度有待持续提高，***可考虑通过短信号召老客回访、平台促销活动、每日签到激励等方式提高用户活跃度***。



仪表盘截图

- SQL**代码：**

```sql
-- 创建PV/UV聚合表
CREATE TABLE pv_uv_puv (
    dates CHAR(10),
    pv INT,
    uv INT,
    puv DECIMAL(10,1)
);

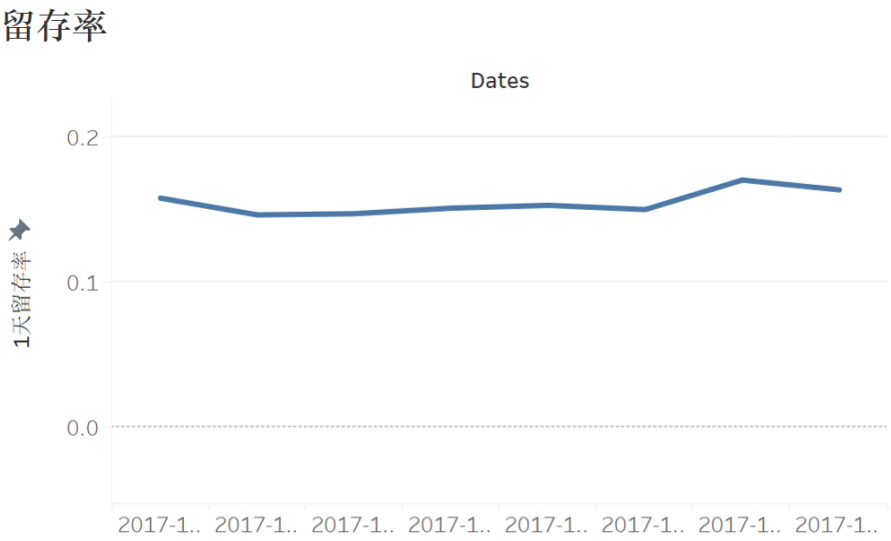
INSERT INTO pv_uv_puv
SELECT
    dates,
    COUNT(IF(behavior_type='pv',1,NULL)) AS pv,
    COUNT(DISTINCT user_id) AS uv,
    ROUND(COUNT(*)/COUNT(DISTINCT user_id),1) AS puv
FROM userbehavior_sample
WHERE behavior_type = 'pv'
GROUP BY dates;
```

二、用户运营体系

2.1 用户留存体系（增长视角）

(1) 留存率矩阵

- **数据结果：**
  - **用户黏性较弱：**用户次日留存率较低，稳定在0.15左右，表明用户黏性较弱，可能与新用户缺乏引导和激励、商品内容匹配程度差有关，***可考虑加强场景化激励、优化内容匹配模型、推行社交化留存等策略***。



- **SQL代码：**

```sql
-- 计算次日/7日留存
SELECT
    a.dates,
    COUNT(DISTINCT b.user_id)/COUNT(DISTINCT a.user_id) AS retention_rate
FROM (
    SELECT DISTINCT user_id, dates
    FROM userbehavior_sample
    WHERE behavior_type = 'pv'
) a
LEFT JOIN (
    SELECT DISTINCT user_id, dates
    FROM userbehavior_sample
    WHERE behavior_type = 'pv'
) b
ON a.user_id = b.user_id
   AND b.dates BETWEEN a.dates + INTERVAL 1 DAY AND a.dates + INTERVAL 7 DAY
GROUP BY a.dates;
```

2.2 用户行为路径（转化视角）

(1) 行为路径标准化处理

- **数据结果：**
  - **用户决策时间长：**90%以上用户行为为直接购买，可能是统计期间较短，未统计到用户前期浏览加购等行为导致，由此可知用户决策时间较长，***可考虑加强用户触达营销、推出限时激励策略等促进购买的最终转化***。

- **购物车/收藏触点未被用户重点使用：**浏览后直接购买为统计期间内主要的用户行为路径，***可考虑优化加购、收藏转化触点。***
- **购物车/收藏触点转化效果较弱：**浏览后加购/收藏再进行购买的转化数量较少，***可考虑加强购物车/收藏营销，如满减提醒、定向优惠券等促进转化。***

用户行为路径汇总

Description	Quantity
直接购买	19866
浏览收藏后购买	1
浏览后购买	394
浏览加购后购买	1
收藏后购买	13
加购后购买	30

行为路径概览

User Id	Item Id	Description	已加购	已收藏	已浏览	已购买
1	4092065	未购买	0.000	0.000	1.000	0.000
	4954999	未购买	0.000	0.000	1.000	0.000
3	1888000	未购买	1.000	0.000	0.000	0.000
4	2102225	未购买	0.000	0.000	1.000	0.000
	3317042	未购买	0.000	0.000	1.000	0.000
	4275344	未购买	0.000	0.000	1.000	0.000
5	5108431	未购买	0.000	0.000	1.000	0.000
6	17756	未购买	0.000	0.000	1.000	0.000
	4160838	未购买	0.000	0.000	1.000	0.000
7	875037	未购买	0.000	0.000	1.000	0.000
	1934023	未购买	0.000	0.000	1.000	0.000
8	543871	未购买	0.000	0.000	1.000	0.000
	1027109	未购买	0.000	0.000	1.000	0.000
10	3135895	未购买	0.000	0.000	1.000	0.000

- **SQL代码：**

```sql
-- 将行为序列编码为二进制状态
CREATE VIEW user_behavior_flag AS
SELECT
    user_id,
    item_id,
    MAX(CASE WHEN behavior_type='pv' THEN 1 ELSE 0 END) AS is_pv,
    MAX(CASE WHEN behavior_type='fav' THEN 1 ELSE 0 END) AS is_fav,
    MAX(CASE WHEN behavior_type='cart' THEN 1 ELSE 0 END) AS is_cart,
    MAX(CASE WHEN behavior_type='buy' THEN 1 ELSE 0 END) AS is_buy
FROM userbehavior_sample
GROUP BY user_id, item_id;

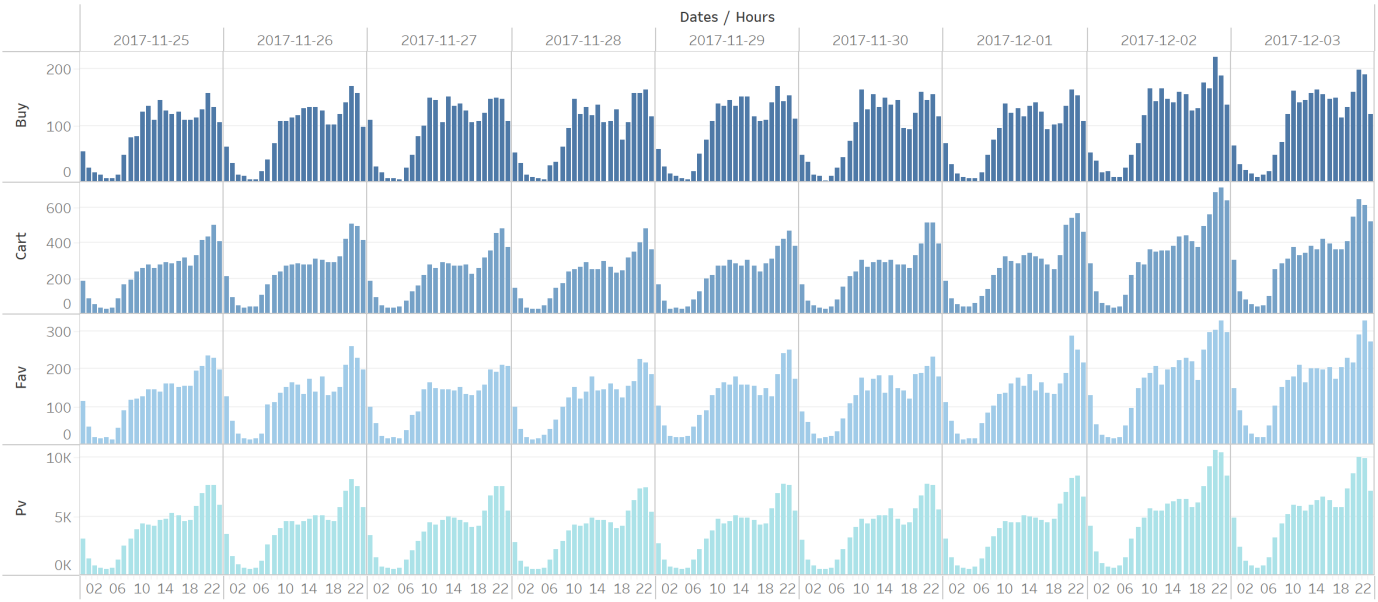
-- 统计典型路径
SELECT
    CONCAT(is_pv, is_fav, is_cart, is_buy) AS path_code,
    COUNT(*) AS user_count
FROM user_behavior_flag
GROUP BY path_code
ORDER BY user_count DESC;
```

(2) 时间序列分析

- **数据结果：**
  - **用户行为集中于中午、夜晚：**整体来看，用户行为主要集中在中午（13时）与夜晚（20时以后），高峰为夜晚21时前后。***可考虑分时段运营策略，午间主推高时效高转化商品，夜间深耕高决策的深度需求产品并叠加限时折扣，以高效促进购买；***

- **不同用户行为时间分布存在一定差异：**具体来看，前期浏览、收藏和加购行为均集中在夜晚21点前后，表明用户习惯于夜晚休息时间进行选购。购买行为高峰分布在中午13时和夜晚21时前后，表明用户习惯于中午与夜晚休息时间进行最终购买。***可考虑断点续购策略，对于中午未完成订单，晚间推送定向提醒以降低订单流失***

用户行为时间序列



- **SQL代码：**

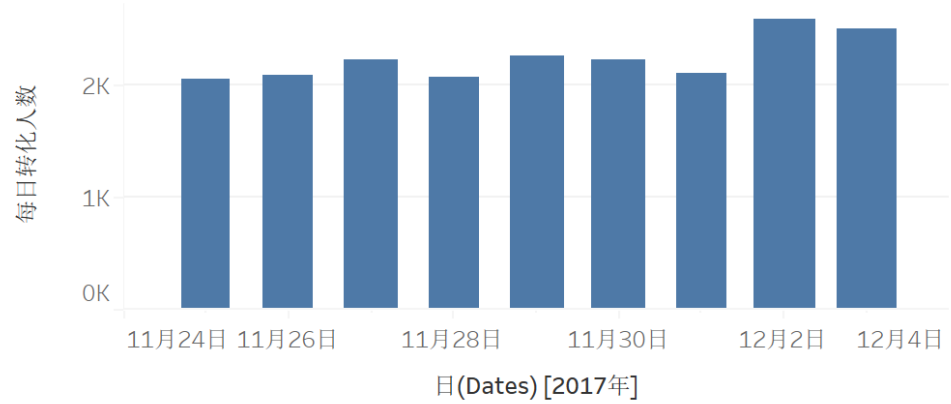
```sql
SELECT
    dates,
    hours,
    COUNT(IF(behavior_type = 'pv', behavior_type, NULL)) AS pv,
    COUNT(IF(behavior_type = 'cart', behavior_type, NULL)) AS cart,
    COUNT(IF(behavior_type = 'fav', behavior_type, NULL)) AS fav,
    COUNT(IF(behavior_type = 'buy', behavior_type, NULL)) AS buy
FROM
    userbehavior_sample
GROUP BY
    dates,
    hours
ORDER BY
    dates,
    hours;
```

(3) 购买行为指标

- **数据结果：**
  - **每日购买转化人数整体呈上升趋势：**受双十二活动激励影响，每日购买转化人数整体呈上升趋势，12月2日后明显增长

- **人均购买频次较少：**人均购买频次整体稳定在1.01，***可考虑加强满减、随单购等活动营销，或优化猜你喜欢等个性化推荐来提频***

购买总人数



购买频次/人

日(Dates)	人均购买..
2017年11月25日	1.01
2017年11月26日	1.01
2017年11月27日	1.01
2017年11月28日	1.01
2017年11月29日	1.01
2017年11月30日	1.01
2017年12月1日	1.01
2017年12月2日	1
2017年12月3日	1.01

- **SQL代码：**

```sql
-- MTU
CREATE TABLE daily_ordering_users (dates DATE, daily_ordering_users_count INT);
INSERT INTO daily_ordering_users (dates, daily_ordering_users_count)
SELECT dates, COUNT(DISTINCT user_id)
FROM userbehavior_sample
WHERE behavior_type = 'buy'
GROUP BY dates
ORDER BY dates;

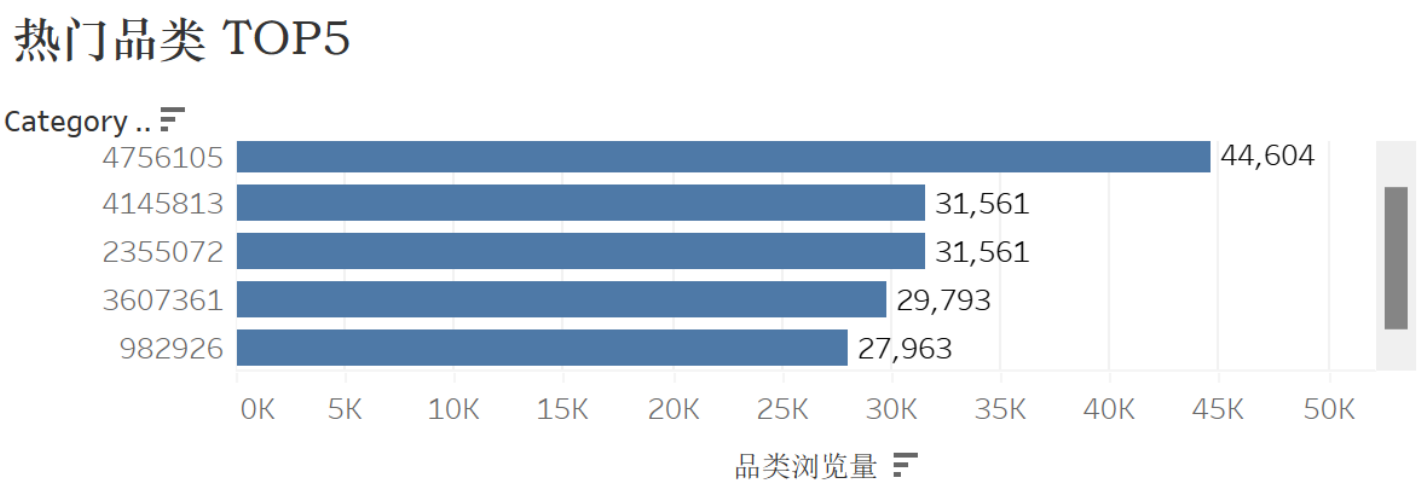
-- 每人每天购买频次
CREATE TABLE daily_avg_purchase_frequency_simplified (dates DATE, daily_avg_frequency DECIMAL(10, 2));
INSERT INTO daily_avg_purchase_frequency_simplified (dates, daily_avg_frequency)
SELECT dates, ROUND(AVG(user_purchase_count), 2)
FROM (
    SELECT dates, user_id, COUNT(*) AS user_purchase_count
    FROM userbehavior_sample
    WHERE behavior_type = 'buy'
    GROUP BY dates, user_id
) AS subquery
GROUP BY dates
ORDER BY dates;
```

三、商品运营分析体系

3.1 商品热度分析

(1) 品类维度

- **数据结果：**
  - 根据热门品类数据结果，***可考虑Top5品类增加专属流量入口，长尾品类采用商品捆绑销售***



- **SQL代码：**

```sql
-- 热门品类Top5
CREATE TABLE hot_categories AS
SELECT
    category_id,
    COUNT(IF(behavior_type='pv',1,NULL)) AS pv
FROM userbehavior_sample
GROUP BY category_id
ORDER BY pv DESC
LIMIT 5;
```

(2) 商品维度

- **数据结果：**
  - 根据热门、热销商品结果，***可考虑采用爆款商品置顶推荐位，低曝光商品优化标题关键词等运营策略***

热销商品 TOP20

Item Id	Sales Volume
257772	6.000
740947	8.000
1034594	9.000
1095113	6.000
1224652	8.000
2560262	6.000
2974129	6.000

热门品类商品 TOP10

Category Id	Item Id	浏览量
1080785	2032668	184
1859277	3371523	192
2355072	640975	164
	987143	164
3607361	138964	210
	2331370	203
	3845720	254

- **SQL代码：**

```sql
-- 商品级PV排行
CREATE TABLE hot_items AS
SELECT
    item_id,
    COUNT(IF(behavior_type='pv',1,NULL)) AS pv
FROM userbehavior_sample
GROUP BY item_id
ORDER BY pv DESC
LIMIT 10;
```

3.2 商品转化漏斗

- **数据结果：**
  - 对于高转化商品，***可考虑优化库存备货***；对于低转化商品，而***可考虑优化详情页设计***
  - 通过分析商品各个环节的点击量，可识别特定商品转化过程中的卡点以优化购买转化结果

购买转化率漏斗

Item Id	Pv	Fav	Cart	Buy	购买转化率
5162808	0.0	0.0	0.0	1.0	1.0
5162729	0.0	0.0	0.0	1.0	1.0
5160873	0.0	0.0	0.0	1.0	1.0
5160870	0.0	0.0	0.0	1.0	1.0
5160743	0.0	0.0	0.0	1.0	1.0
5159189	0.0	0.0	0.0	1.0	1.0
5158534	0.0	0.0	0.0	1.0	1.0
5157959	0.0	0.0	0.0	1.0	1.0
5154814	1.0	0.0	0.0	1.0	1.0
5154390	0.0	0.0	0.0	1.0	1.0
5153171	0.0	0.0	0.0	1.0	1.0
5151396	1.0	0.0	0.0	1.0	1.0
5150587	0.0	0.0	0.0	1.0	1.0

- **SQL代码：**

```sql
-- 构建商品转化分析表
CREATE TABLE item_conversion (
    item_id INT,
    pv INT,
    fav INT,
    cart INT,
    buy INT,
    conversion_rate FLOAT
);

INSERT INTO item_conversion
SELECT
    item_id,
    COUNT(IF(behavior_type='pv',1,NULL)) AS pv,
    COUNT(IF(behavior_type='fav',1,NULL)) AS fav,
    COUNT(IF(behavior_type='cart',1,NULL)) AS cart,
    COUNT(IF(behavior_type='buy',1,NULL)) AS buy,

COUNT(IF(behavior_type='buy',1,NULL))/COUNT(IF(behavior_type='pv',1,NULL)) AS
conversion_rate
FROM userbehavior_sample
GROUP BY item_id;
```

Appendix

数据预处理

1.1 主键去重（解决重复点击问题）

业务背景：同一用户对同一商品在同一秒可能产生多次行为记录（如连续点击）

```sql
-- 通过(user_id, item_id, timestamps)三字段组合去重
DELETE FROM userbehavior_sample
WHERE id NOT IN (
    SELECT MIN(id)  -- 保留最早行为记录
    FROM userbehavior_sample
    GROUP BY user_id, item_id, timestamps
    HAVING COUNT(*) > 1
);
```

实现效果：

- 消除重复记录对UV计算的干扰
- 确保用户行为序列的时序准确性

1.2 时间维度拆解（支撑时段分析）

业务需求：按日/小时分析流量规律，定位高峰时段

```sql
-- 从时间戳提取完整时间维度
ALTER TABLE userbehavior_sample
ADD COLUMN datetimes TIMESTAMP(0) GENERATED ALWAYS AS
(FROM_UNIXTIME(timestamps)),
ADD COLUMN dates CHAR(10) GENERATED ALWAYS AS (SUBSTRING(datetimes, 1, 10)),
ADD COLUMN hours CHAR(2) GENERATED ALWAYS AS (SUBSTRING(datetimes, 12, 2));
```

字段用途：

字段名	示例值	分析场景
dates	2017-12-01	日趋势分析（如双12流量监控）
hours	20	小时级流量波动（晚高峰识别）

1.3 异常数据清洗（限定分析周期）

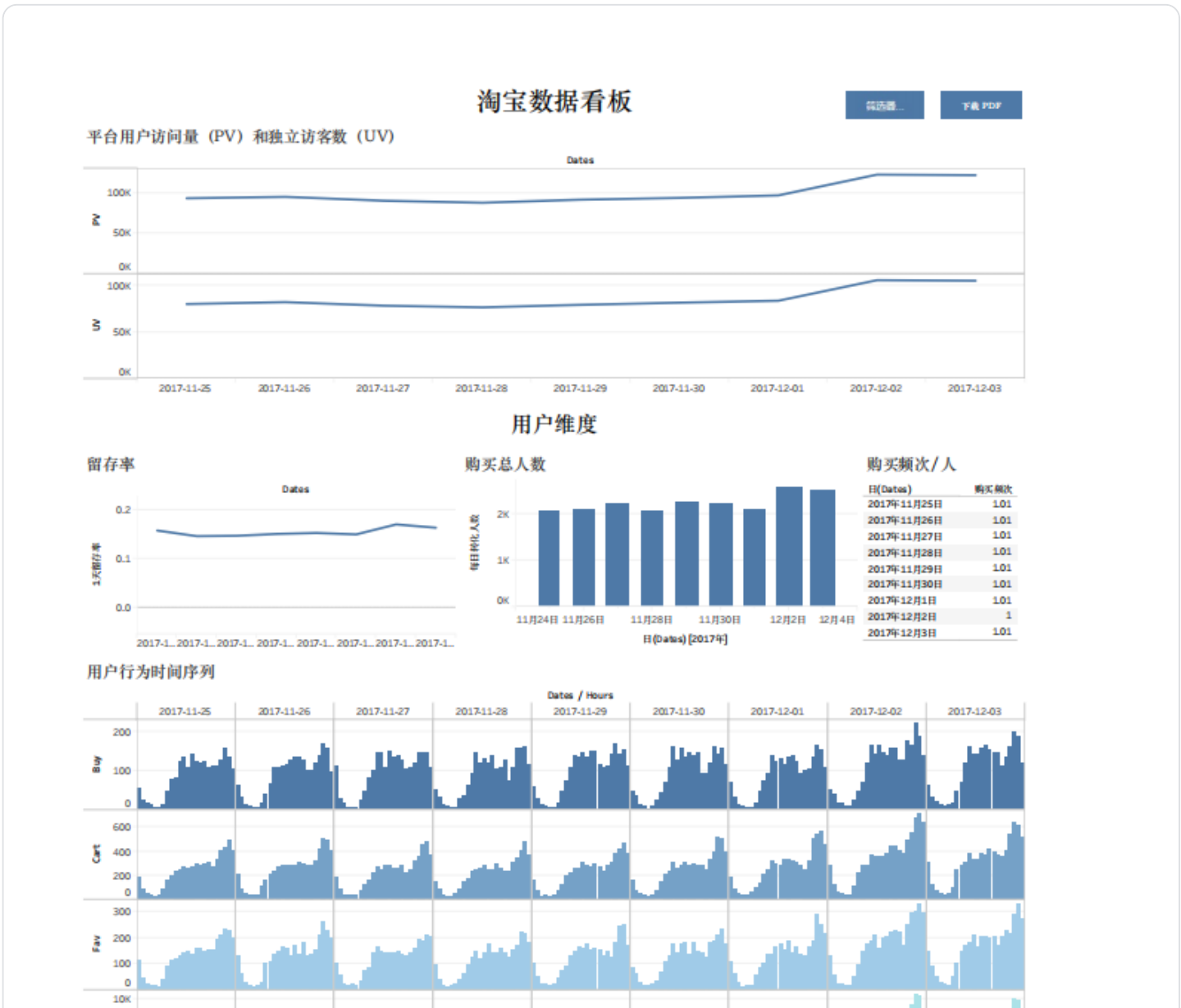
数据规则：限定分析周期为2017-11-25至2017-12-03

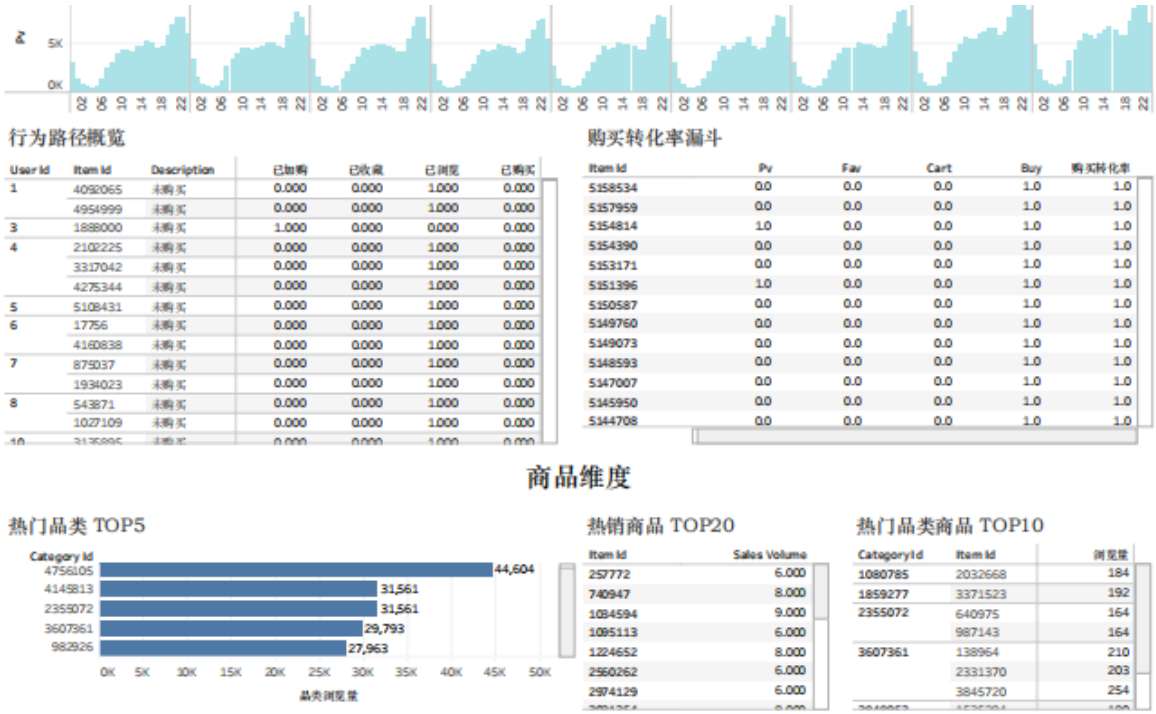
```sql
-- 删除超出时间范围的记录
DELETE FROM userbehavior_sample
WHERE dates < '2017-11-25' OR dates > '2017-12-03';
```

处理结果：

- 有效数据量：999,459条

Tableau仪表盘截图





淘宝数据看板.pdf

数据处理源代码

```
Part1《数据预处理》

-- 去重
alter table userbehavior_sample MODIFY id INT PRIMARY KEY auto_increment;
DELETE userbehavior_sample from
userbehavior_sample,
(
SELECT user_id, item_id, timestamps, MIN(id) id from userbehavior_sample
GROUP BY user_id, item_id, timestamps
HAVING count(*)>1
) as t2
WHERE userbehavior_sample.user_id=t2.user_id
and userbehavior_sample.item_id=t2.item_id
and userbehavior_sample.timestamps=t2.timestamps
and userbehavior_sample.id=t2.id

-- 新增日期: date time hour
-- 更改buffer值
show VARIABLES like '%_buffer%'
```

```
set global innodb_buffer_pool_size=10000000000
-- datetime
ALTER table userbehavior_sample add datetimes TIMESTAMP(0);
update userbehavior_sample set datetimes=FROM_UNIXTIME(timestamps);
-- date
ALTER table userbehavior_sample add dates char(10);
ALTER table userbehavior_sample add times char(8);
ALTER table userbehavior_sample add hours char(8);
update userbehavior_sample set dates=SUBSTRING(datetimes,1,10);
update userbehavior_sample set times=SUBSTRING(datetimes,12,8);
update userbehavior_sample set hours=SUBSTRING(datetimes,12,2);
SELECT * from userbehavior_sample limit 5

-- 去异常
SELECT MAX(datetimes), MIN(datetimes) from userbehavior_sample;
DELETE from userbehavior_sample
WHERE datetimes<'2017-11-25 00:00:00'
or datetimes>'2017-12-03 23:59:59'
DELETE from userbehavior_sample WHERE dates is NULL

-- 数据概览
SELECT COUNT(1) from userbehavior_sample


Part2 数据分析

SELECT * FROM `userbehavior_sample`

-- 创建临时表
create table temp_behavior like userbehavior_sample
-- 截取数据
INSERT INTO temp_behavior
SELECT * from userbehavior_sample limit 100000;
-- 测试
SELECT * from temp_behavior;

-- 获客情况
-- 页面浏览量（pv）
SELECT dates, count('pv') as pv from temp_behavior
where behavior_type='pv'
GROUP BY dates
ORDER BY dates;
-- 独立访客数
SELECT dates, count(DISTINCT user_id) as unique_user from temp_behavior
where behavior_type='pv'
GROUP BY dates
ORDER BY dates;
```

```sql
-- 合并
SELECT dates, count('pv') as pv, count(DISTINCT user_id) as unique_user,
ROUND(count('pv')/count(DISTINCT user_id),1)  as 'pv/uv' from temp_behavior
where behavior_type='pv'
GROUP BY dates
ORDER BY dates;

-- 处理真实数据
CREATE table pv_uv_puv (
dates char(10),
pv int(9),
uv int(9),
puv DECIMAL(10,1)
);

INSERT into pv_uv_puv
SELECT dates, count('pv') as pv, count(DISTINCT user_id) as unique_user,
ROUND(count('pv')/count(DISTINCT user_id),1)  as 'pv/uv' from us
where behavior_type='pv'
GROUP BY dates
ORDER BY dates;

SELECT * from pv_uv_puv
DELETE from pv_uv_puv WHERE dates is NULL

SELECT * FROM userbehavior_sample limit 5

-- 计算DAU、MAU
CREATE TABLE DAU(
dates char(10),
DAU int(6)
);
INSERT INTO DAU
SELECT dates, COUNT(DISTINCT user_id) AS DAU
FROM userbehavior_sample
WHERE behavior_type = 'pv'
GROUP BY dates;

SELECT COUNT(DISTINCT user_id) AS user_count
FROM userbehavior_sample
WHERE behavior_type = 'pv'
  AND dates BETWEEN '2017-11-25' AND '2017-12-03';


-- 留存数（自关联）
SELECT
    a.dates,
```

```sql
    COUNT(IF(DATEDIFF(b.dates, a.dates) = 0, b.user_id, NULL)) AS retention_0,
    COUNT(IF(DATEDIFF(b.dates, a.dates) = 1, b.user_id, NULL)) AS retention_1,
    COUNT(IF(DATEDIFF(b.dates, a.dates) = 7, b.user_id, NULL)) AS retention_7
FROM
    (SELECT user_id, dates
     FROM userbehavior_sample
     GROUP BY user_id, dates
    ) a
JOIN
    (SELECT user_id, dates
     FROM userbehavior_sample
     GROUP BY user_id, dates
    ) b ON a.user_id = b.user_id AND a.dates <= b.dates
GROUP BY a.dates;

-- 留存率
SELECT
    a.dates,
    COUNT(IF(DATEDIFF(b.dates, a.dates) = 1, b.user_id,
NULL))/COUNT(IF(DATEDIFF(b.dates, a.dates) = 0, b.user_id, NULL)) AS
retention_rate_1,
    COUNT(IF(DATEDIFF(b.dates, a.dates) = 7, b.user_id, NULL))
/COUNT(IF(DATEDIFF(b.dates, a.dates) = 0, b.user_id, NULL)) AS
retention_rate_7
FROM
    (SELECT user_id, dates
     FROM userbehavior_sample
     GROUP BY user_id, dates
    ) a
JOIN
    (SELECT user_id, dates
     FROM userbehavior_sample
     GROUP BY user_id, dates
    ) b ON a.user_id = b.user_id AND a.dates <= b.dates
GROUP BY a.dates
order by a.dates;

-- 时间序列（周内&日内行为分析）
SELECT
    dates,
    hours,
    COUNT(IF(behavior_type = 'pv', behavior_type, NULL)) AS pv,
    COUNT(IF(behavior_type = 'cart', behavior_type, NULL)) AS cart,
    COUNT(IF(behavior_type = 'fav', behavior_type, NULL)) AS fav,
    COUNT(IF(behavior_type = 'buy', behavior_type, NULL)) AS buy
FROM
    userbehavior_sample
```

```sql
GROUP BY
    dates,
    hours
ORDER BY
    dates,
    hours;

-- 创建表
CREATE TABLE date_hour_behavior (
    dates CHAR(10),
    hours CHAR(2),
    pv INT,
    cart INT,
    fav INT,
    buy INT
);

INSERT INTO date_hour_behavior
SELECT
    dates,
    hours,
    COUNT(IF(behavior_type = 'pv', behavior_type, NULL)) AS pv,
    COUNT(IF(behavior_type = 'cart', behavior_type, NULL)) AS cart,
    COUNT(IF(behavior_type = 'fav', behavior_type, NULL)) AS fav,
    COUNT(IF(behavior_type = 'buy', behavior_type, NULL)) AS buy
FROM
    userbehavior_sample
GROUP BY
    dates,
    hours
ORDER BY
    dates,
    hours;

-- 统计各类行为用户数
SELECT behavior_type,
COUNT(DISTINCT user_id) user_number
from userbehavior_sample
GROUP BY behavior_type
ORDER BY behavior_type DESC

-- 创建表
CREATE TABLE behavior_number(
behavior_type VARCHAR(5),
user_number int);

INSERT INTO behavior_number
```

```sql
SELECT behavior_type,
COUNT(DISTINCT user_id) user_number
from userbehavior_sample
GROUP BY behavior_type
ORDER BY behavior_type DESC

-- 统计各类行为数
CREATE TABLE behavior_number_0(
behavior_type VARCHAR(5),
user_number int);

INSERT INTO behavior_number_0
SELECT behavior_type,
COUNT(*) user_number
from userbehavior_sample
GROUP BY behavior_type
ORDER BY behavior_type DESC

SELECT (28586+55248)/895308 --收藏和加购转化率
SELECT 20317/895308 --购买转化率

-- 用户行为路径分析
CREATE VIEW user_behavior_view AS
SELECT user_id, item_id,
COUNT(if(behavior_type='pv',behavior_type,NULL)) pv,
COUNT(if(behavior_type='fav',behavior_type,NULL))fav,
COUNT(if(behavior_type='cart',behavior_type,NULL)) cart,
COUNT(if(behavior_type='buy',behavior_type,NULL))buy
FROM userbehavior_sample
GROUP BY user_id, item_id;

-- 用户行为标准化
CREATE VIEW user_behavior_standard AS
SELECT user_id, item_id,
(CASE WHEN pv>0 THEN 1 ELSE 0 END ) '已浏览',
(CASE WHEN fav>0 THEN 1 ELSE 0 END ) '已收藏',
(CASE WHEN cart>0 THEN 1 ELSE 0 END ) '已加购',
(CASE WHEN buy>0 THEN 1 ELSE 0 END ) '已购买'
FROM user_behavior_view;

-- 路径类型
CREATE VIEW user_behavior_path AS
SELECT *, CONCAT(已浏览, 已收藏, 已加购, 已购买) '购买路径类型'
from user_behavior_standard as a
where a.已购买>0


```

```sql
-- 统计各类购买行为数量
CREATE VIEW path_count AS
SELECT 购买路径类型, count(*) 数量 from user_behavior_path
GROUP BY 购买路径类型
order by 数量
SELECT * from path_count


CREATE table description(
path_type char(4),
description VARCHAR(40));

INSERT into description
VALUES('0001','直接购买'),
('1001','浏览后购买'),
('0011','加购后购买'),
('1011','浏览加购后购买'),
('0101','收藏后购买'),
('1101','浏览收藏后购买'),
('0111','收藏加购后购买'),
('1111','浏览收藏加购后购买')


-- 合并standard和description
CREATE TABLE user_behavior_standard_table
AS
SELECT *, CONCAT(已浏览, 已收藏, 已加购, 已购买) '购买路径类型' FROM
user_behavior_standard;
SELECT * from  user_behavior_standard_table LIMIT 5


CREATE TABLE user_behavior_combined_standard (
    user_id INT,
    item_id INT,
    `已浏览` INT,
    `已收藏` INT,
    `已加购` INT,
    `已购买` INT,
    `购买路径类型` VARCHAR(10),
    description VARCHAR(40)
);
INSERT INTO user_behavior_combined_standard (user_id, item_id, `已浏览`, `已收
藏`, `已加购`, `已购买`, `购买路径类型`, description)
SELECT
    a.user_id,
    a.item_id,
    a.`已浏览`,
```

```
    a.`已收藏`,
    a.`已加购`,
    a.`已购买`,
    a.`购买路径类型`,
    d.description
FROM
    user_behavior_standard_table a
LEFT JOIN
    description d ON a.`购买路径类型` = d.path_type;

UPDATE user_behavior_combined_standard
SET description = '未购买'
WHERE description IS NULL;


SELECT * from path_count p
JOIN description d
on p.购买路径类型=d.path_type
ORDER BY 数量 desc

drop VIEW user_behavior_view;
drop VIEW user_behavior_standard;
drop VIEW user_behavior_path;
drop VIEW path_count;

-- 创建表
CREATE table path_result(
path_type char(4),
description varchar(40),
num int);

CREATE VIEW user_behavior_view AS
SELECT user_id, item_id,
COUNT(if(behavior_type='pv',behavior_type,NULL)) pv,
COUNT(if(behavior_type='fav',behavior_type,NULL))fav,
COUNT(if(behavior_type='cart',behavior_type,NULL)) cart,
COUNT(if(behavior_type='buy',behavior_type,NULL))buy
FROM userbehavior_sample
GROUP BY user_id, item_id;

-- 用户行为标准化
CREATE VIEW user_behavior_standard AS
SELECT user_id, item_id,
(CASE WHEN pv>0 THEN 1 ELSE 0 END ) '已浏览',
(CASE WHEN fav>0 THEN 1 ELSE 0 END ) '已收藏',
(CASE WHEN cart>0 THEN 1 ELSE 0 END ) '已加购',
(CASE WHEN buy>0 THEN 1 ELSE 0 END ) '已购买'
```

```sql
FROM user_behavior_view;

-- 路径类型
CREATE VIEW user_behavior_path AS
SELECT *, CONCAT(已浏览, 已收藏, 已加购, 已购买) '购买路径类型'
from user_behavior_standard as a
where a.已购买>0

-- 统计各类购买行为数量
CREATE VIEW path_count AS
SELECT 购买路径类型, count(*) 数量 from user_behavior_path
GROUP BY 购买路径类型
order by 数量


INSERT into path_result
SELECT path_type,description,数量 from path_count p
JOIN description d
on p.购买路径类型=d.path_type
ORDER BY 数量 desc

SELECT * from path_result

-- MTU
CREATE TABLE daily_ordering_users (dates DATE, daily_ordering_users_count INT);
INSERT INTO daily_ordering_users (dates, daily_ordering_users_count)
SELECT dates, COUNT(DISTINCT user_id)
FROM userbehavior_sample
WHERE behavior_type = 'buy'
GROUP BY dates
ORDER BY dates;

-- 每人每天购买频次
CREATE TABLE daily_avg_purchase_frequency_simplified (dates DATE, daily_avg_frequency DECIMAL(10, 2));
INSERT INTO daily_avg_purchase_frequency_simplified (dates, daily_avg_frequency)
SELECT dates, ROUND(AVG(user_purchase_count), 2)
FROM (
    SELECT dates, user_id, COUNT(*) AS user_purchase_count
    FROM userbehavior_sample
    WHERE behavior_type = 'buy'
    GROUP BY dates, user_id
) AS subquery
GROUP BY dates
ORDER BY dates;
```

```sql

-- 热门品类
SELECT category_id,
count(if(behavior_type='pv', behavior_type, null) )'品类浏览量'
from userbehavior_sample
group by category_id
ORDER BY 2 DESC
limit 5

-- 热门商品
SELECT item_id,
count(if(behavior_type='pv', behavior_type, null) )'商品浏览量'
from userbehavior_sample
group by item_id
ORDER BY 2 DESC
limit 5

--热门品类中的热门商品
SELECT category_id,item_id,品类商品浏览量 from(
SELECT category_id,item_id,
count(if(behavior_type='pv', behavior_type, null) )'品类商品浏览量',
rank()over(PARTITION by category_id ORDER BY '品类商品浏览量' DESC) r
from userbehavior_sample
group by category_id,item_id
ORDER BY 3 DESC) a
where r=1
order by a.品类商品浏览量 DESC
limit 10

-- 创建表存储
CREATE TABLE popular_categroies(
category_id int,
品类浏览量 int);
INSERT into popular_categroies
SELECT category_id,
count(if(behavior_type='pv', behavior_type, null) )'品类浏览量'
from userbehavior_sample
group by category_id
ORDER BY 2 DESC
limit 5

CREATE TABLE popular_item(
item_id int,
商品浏览量 int);
INSERT INTO popular_item
SELECT item_id,
count(if(behavior_type='pv', behavior_type, null) )'商品浏览量'
```

```sql
from userbehavior_sample
group by item_id
ORDER BY 2 DESC
limit 5

CREATE TABLE popular_category_item(
category_id int,
item_id int,
品类商品浏览量 int);
INSERT into popular_category_item
SELECT category_id,item_id,品类商品浏览量 from(
SELECT category_id,item_id,
count(if(behavior_type='pv', behavior_type, null) )'品类商品浏览量',
rank()over(PARTITION by category_id ORDER BY '品类商品浏览量' DESC) r
from userbehavior_sample
group by category_id,item_id
ORDER BY 3 DESC) a
where r=1
order by a.品类商品浏览量 DESC
limit 10


CREATE TABLE popular_sale_item (
    item_id INT,
    sales_volume INT
);
INSERT INTO popular_sale_item (item_id, sales_volume)
SELECT
    item_id,
    COUNT(IF(behavior_type = 'buy', behavior_type, NULL)) AS sales_volume
FROM
    userbehavior_sample
GROUP BY
    item_id
ORDER BY 2 DESC
limit 20;


-- 商品转化率
CREATE TABLE        item_detail(
item_id int,
pv int,
fav int,
cart int,
buy int,
item_transction_rate float);

```

```sql
INSERT into item_detail
SELECT
    item_id,
    COUNT(IF(behavior_type = 'pv', behavior_type, NULL)) AS pv,
    COUNT(IF(behavior_type = 'fav', behavior_type, NULL)) AS fav,
    COUNT(IF(behavior_type = 'cart', behavior_type, NULL)) AS cart,
    COUNT(IF(behavior_type = 'buy', behavior_type, NULL)) AS buy,
    COUNT(DISTINCT IF(behavior_type = 'buy', user_id, NULL)) / COUNT(DISTINCT user_id) AS 商品转化率
FROM
    userbehavior_sample
GROUP BY
    item_id
ORDER BY
    商品转化率 DESC;
```